GRADE 7 ACCELERATED · UNIT 5 · LESSONS 10–11

Multi-Step Proportion Problems

Independent Set 5.10 **ANSWER KEY**

Name
Date
Period

Six questions, on your own, in silence. Show the step that does the work.

1. A car gets 32 miles per gallon and its tank holds 14 gal. How far does it go on a tank?

448 mi

2. That car makes a 1,000-mi trip. How many tanks of gas does it use? (Part of a tank counts as a whole tank.)

3 tanks

1,000 ÷ 448 ≈ 2.23 → 3.

3. A recipe for 4 servings uses $1\frac{1}{4}$ cups of rice. How much rice for 10 servings?

3⅛ cups

k = 2.5; 2.5 × 1¼.

4. $1\,\text{km} = 0.62\,\text{mi}$. A train does 120 km/h; a bus does 70 mi/h. Which is faster? Show the speeds in one unit.

The train: 120 km/h = 74.4 mi/h > 70 mi/h

120 × 0.62.

5. Juice is mixed orange : mango : water = $3:2:5$. A batch uses 9 cups of orange. How many cups is the whole batch?

30 cups

k = 3; 10 parts × 3.

6. A plan is drawn at $1\,\text{in} : 6\,\text{ft}$; a room is 2 in by 3.5 in on it. Find the room's real area.

252 ft²

12 ft by 21 ft. (The square-meter version is the test's transfer item and is not practised here — ruling 18.)